

Desigualdad de Jensen

Teorema 1 (Desigualdad de Jensen). Sea (X, Σ, μ) un espacio de medida finito, $\varphi: I \rightarrow \mathbb{R}$ una función convexa en el intervalo $I = (a, b) \subset \mathbb{R}$ y $f: X \rightarrow I$ en $\mathcal{L}^1$

$$\implies \varphi \left(\frac{1}{\mu(X)} \int_X f \, d\mu \right) \leq \frac{1}{\mu(X)} \int_X \varphi(f) \, d\mu.$$

Demostración: Veamos dos casos por separado:

Caso I. Sea (X, Σ, μ) un espacio de probabilidad (i.e. $\mu(X) = 1$). Sea $x_0 := \int_X f \, d\mu$, entonces $a \leq \inf_{x \in X} f(x) \leq x_0 \leq \sup_{x \in X} f(x) \leq b$.

Definimos $\beta := \sup_{a < x < x_0} \frac{\varphi(x_0) - \varphi(x)}{x_0 - x}$. Como φ es convexa, por la **prop-carac-fn-convexa/??**,

$$\forall y \in (x_0, b) : \frac{\varphi(x_0) - \varphi(x)}{x_0 - x} \leq \frac{\varphi(y) - \varphi(x_0)}{y - x_0} \implies \beta \leq \frac{\varphi(y) - \varphi(x_0)}{y - x_0}.$$

Esto significa que $\forall y \in (x_0, b) : \varphi(y) \geq \beta(y - x_0) + \varphi(x_0)$. Por definición de β , también se cumple que $\forall x \in (a, x_0) : \varphi(x) \geq \beta(x - x_0) + \varphi(x_0)$.

$$\implies \forall s \in (a, b) : \varphi(s) \geq \beta(s - x_0) + \varphi(x_0).$$

Tomando $s = f(x)$, e integrando, obtenemos que

$$\begin{aligned} \int_X \varphi(f(x)) \, d\mu(x) &\geq \int_X \beta(f(x) - x_0) \, d\mu(x) + \int_X \varphi(x_0) \, d\mu(x) = \\ &\geq \beta \int_X f(x) \, d\mu(x) - \beta x_0 \mu(X) + \varphi(x_0) \mu(X) = \varphi \left(\int_X f \, d\mu \right). \end{aligned}$$

Caso II. Si $\mu(X) \neq 1$, definimos ν mediante $\forall A \in \Sigma : \nu(A) = \frac{\mu(A)}{\mu(X)}$. Entonces, ν es una medida de probabilidad y, por el caso **??**, tenemos que

$$\varphi \left(\int_X f \, d\nu \right) \leq \int_X \varphi(f) \, d\nu \iff \varphi \left(\frac{1}{\mu(X)} \int_X f \, d\mu \right) \leq \frac{1}{\mu(X)} \int_X \varphi(f) \, d\mu. \quad \blacksquare$$

Referenciado en

- `cor-orden-normas-lp`
- `ejer-desigualdad-aritmetico-geometrica-jensen`
- `prop-inclusion-lp-esp-finito`
- `desigualdad-holder`

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